CS3005: DIGITAL MEDIA AND GAMES

3D Graphics for Games

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Gaming

VFX

VR

Animated Movies

Architectural Visualisations

3D Printing

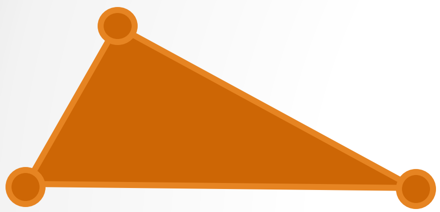
Requires Basic 3D

Requires Basic 3D

3D Graphics Pipeline

Application

↓



Geometry — Geometry Pipeline

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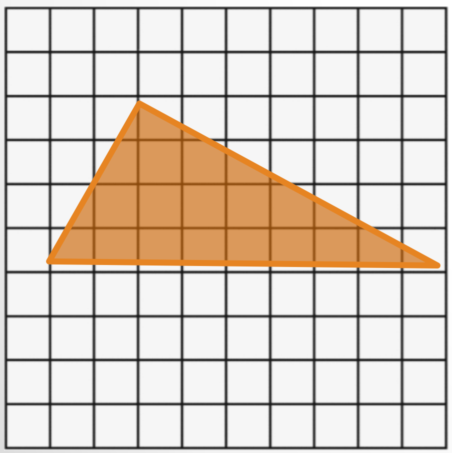


Rasterisation — Rasterise Triangles and Interpolate the Geometry Attributes from Screen Coordinate to Pixel

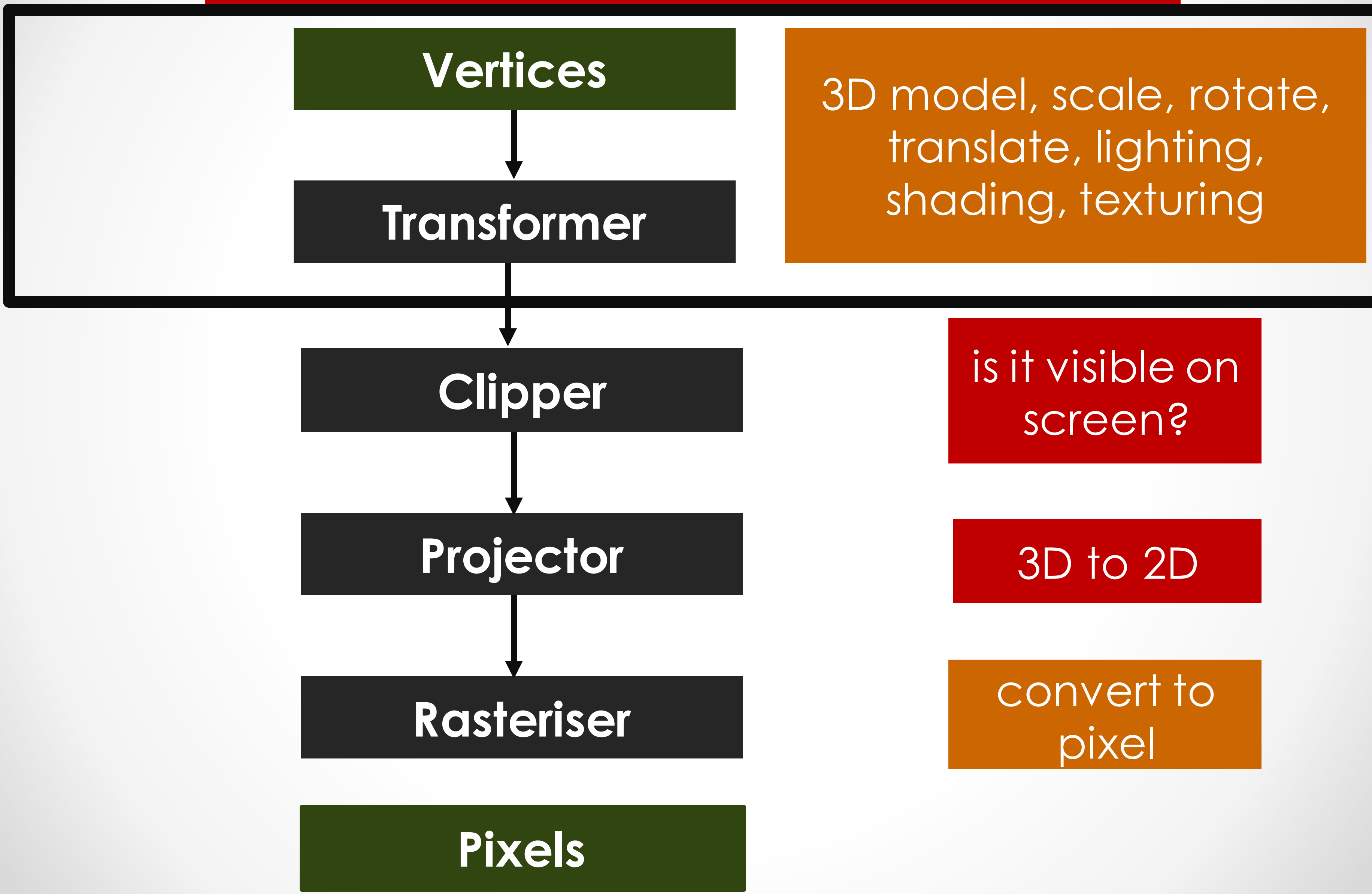
↓

Screen

3D Graphics Pipeline

Vertices	
Transformer	3D model, scale, rotate, translate, lighting, shading, texturing
Clipper	is it visible on screen?
Projector	3D to 2D
Rasteriser	convert to pixel
Pixels	

3D Graphics Pipeline

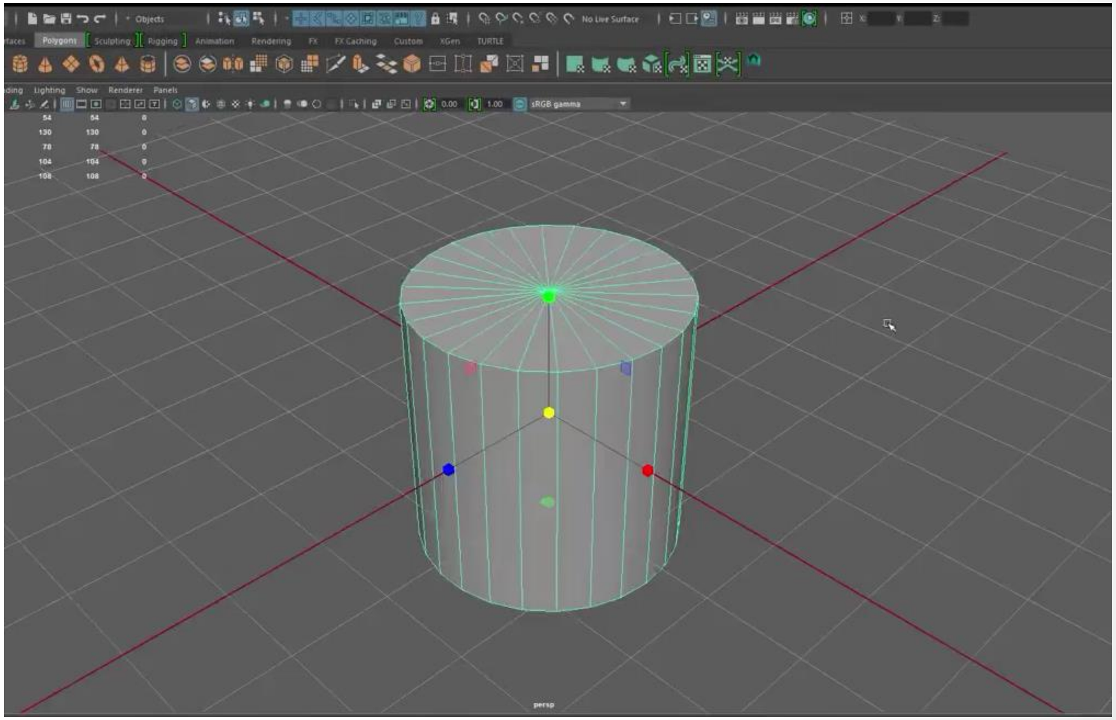


3 Building Blocks of 3D

Modelling

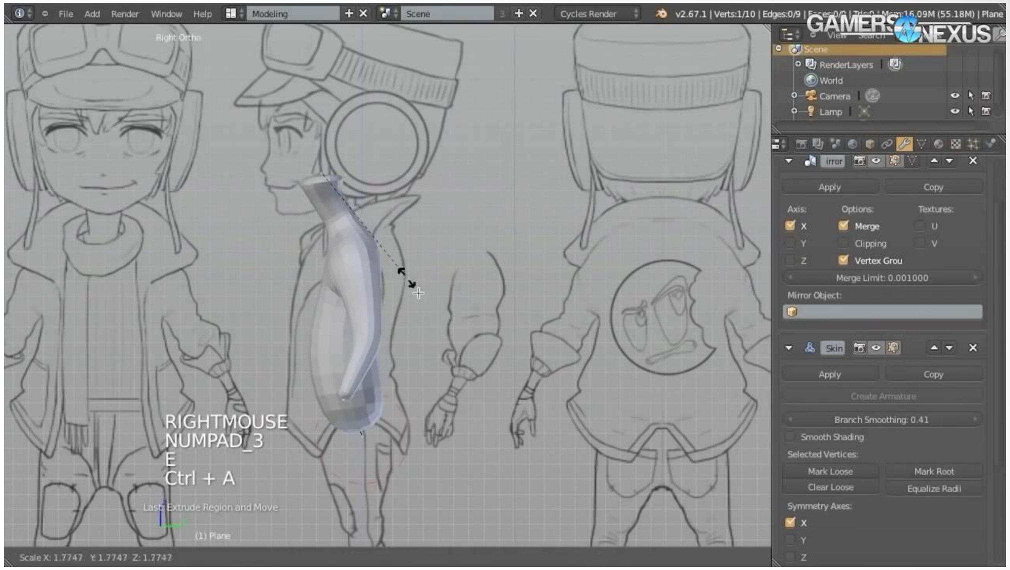
Lighting

Texturing



Modelling

3d Modelling design



3d Capture

3D Geometry

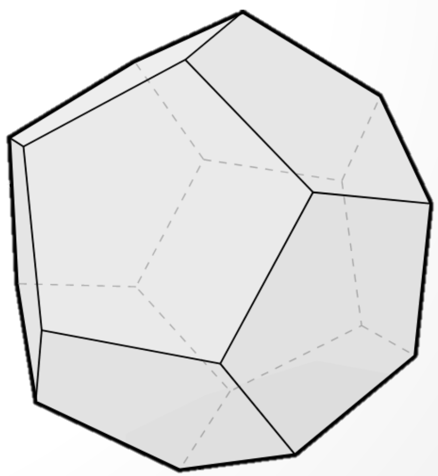
3D Geometry

earth measure

geo • metry :

- The study of shapes, sizes, patterns, and positions.

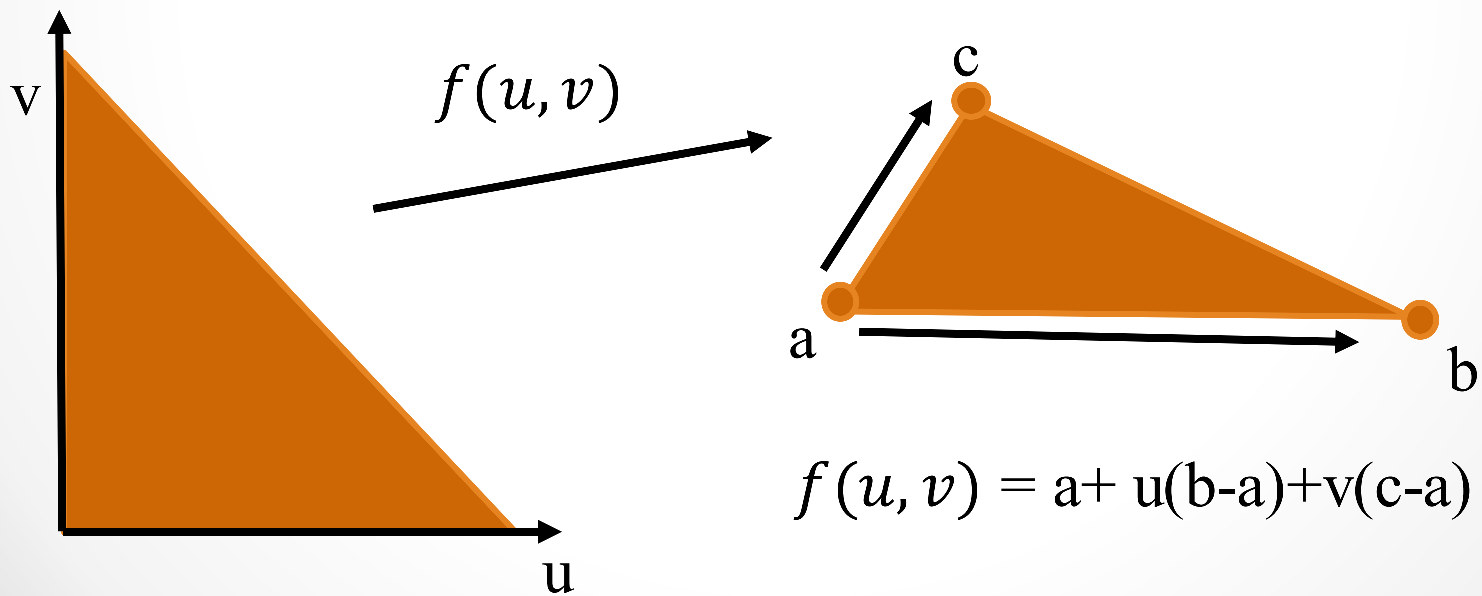
- The study of spaces where some quantity (lengths, angles, etc.) can be measured.



Triangle Mesh

The reason why triangles are commonly used in practice is **simplicity**:

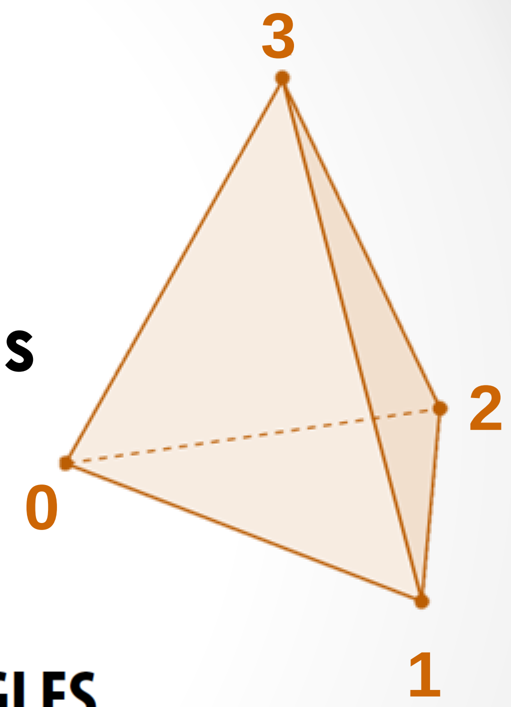
Three points in space unambiguously define a plane



$$f(u,v) = \mathrm{a} + \mathrm{u}(\mathrm{b}-\mathrm{a}) + \mathrm{v}(\mathrm{c}-\mathrm{a})$$

Triangle Mesh

- Store vertices as triples of coordinates (x,y,z)
- Store triangles as triples of indices (i,j,k)
- E.g., tetrahedron



VERTICES

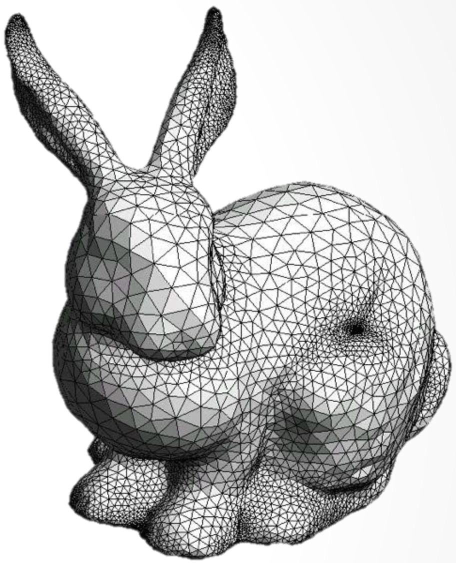
	x	y	z
0:	−1	−1	−1
1:	1	−1	1
2:	1	1	−1
3:	−1	1	1

TRIANGLES

i	j	k
0	2	1
0	3	2
3	0	1
3	1	2

Triangle Mesh

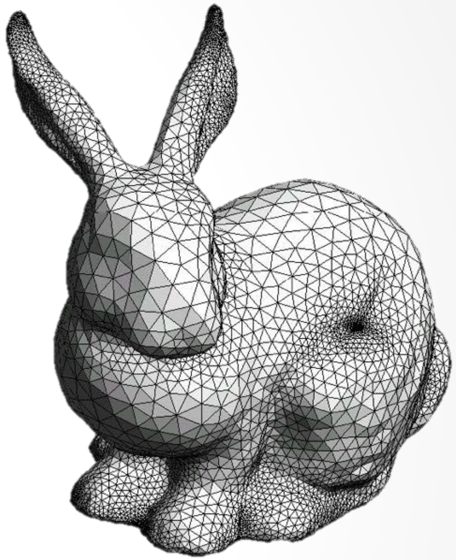
```
# vertex count = 2503
# face count = 4968
v -3.4101800e-003 1.3031957e-001 2.1754370e-002
v -8.1719160e-002 1.5250145e-001 2.9656090e-002
v -3.0543480e-002 1.2477885e-001 1.0983400e-003
v -2.4901590e-002 1.1211138e-001 3.7560240e-002
v -1.8405680e-002 1.7843055e-001 -2.4219580e-002
v 1.9067940e-002 1.2144925e-001 3.1968440e-002
v 6.0412000e-003 1.2494359e-001 3.2652890e-002
v -1.3469030e-002 1.6299355e-001 -1.2000020e-002
v -3.4393240e-002 1.7236688e-001 -9.8213000e-004
v -8.4314160e-002 1.0957263e-001 3.7097300e-003
v -4.2233540e-002 1.7211574e-001 -4.1799800e-003
v -6.3308390e-002 1.5660615e-001 -1.3838790e-002
v -7.6903950e-002 1.6708033e-001 -2.6931360e-002
v -7.2253920e-002 1.1539550e-001 5.1670300e-002
f 420 1071 1065
f 264 281 254
f 298 309 281
f 368 366 367
f 368 396 366
f 1639 1564 1139
f 560 48 47
f 82 471 212
f 25 38 37
f 202 1206 1080
f 264 298 281
f 298 331 309
f 309 331 366
```



OBJ File

Triangle Mesh

Game designers have to carefully balance **model fidelity vs polygon count**, because if the count goes too high, the **framerate** of an animation drops below what users perceive as smooth.



3D Transformations

Translation

Position

Scaling

Size

Rotation

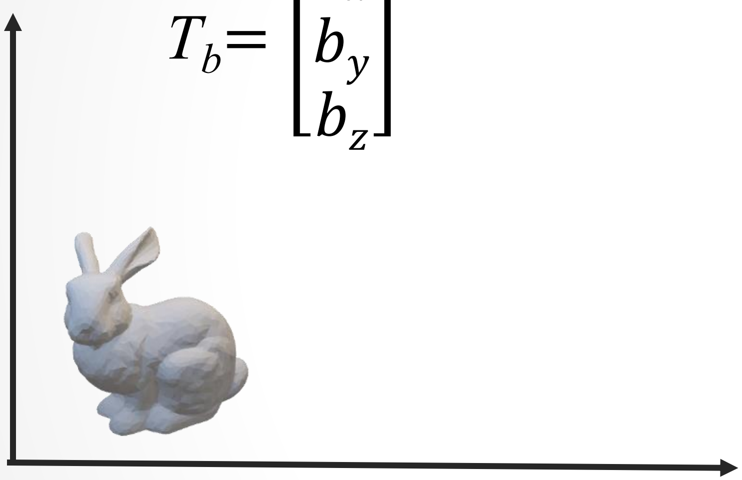
orientation

represent 3D transformations as matrices

3D Transformations

Translation

$$T_b = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}$$



$$\mathrm{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Homogeneous Coordinates

$$\mathrm{T}_b = \begin{bmatrix} \boldsymbol{I} & T_b \\ \boldsymbol{0}^{\boldsymbol{T}} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & b_x \\ 0 & 1 & 0 & by \\ 0 & 0 & 1 & b_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & b_x \\ 0 & 1 & 0 & by \\ 0 & 0 & 1 & b_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\mathrm{v}' = T_b v$$

3D Transformations

Scaling

$$S_s = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & S_z \end{bmatrix}$$

$$\mathrm{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\mathrm{S_s} = \begin{bmatrix} S_s & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogeneous Coordinates

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\mathrm{v}' = S_s v$$

3D Transformations

Rotation

$R_{yz}(\theta)$: rotation about x axis by θ

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogeneous Coordinates

$$\theta = 90°$$

$$v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$v' = R_x v$$

3D Transformations

Rotation

$R_{zx}(\theta)$: rotation about y axis by θ

$$\mathrm{R_y} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogeneous Coordinates

$\theta = 90°$

$$\mathrm{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\mathrm{v}' = R_y v$$

3D Transformations

Rotation

$R_{xy}(\theta)$: rotation about z axis by θ

$$R_z = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Homogeneous Coordinates

$$\theta = 90°$$

$$v = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$v' = R_z v$$

why this is so beautiful ?

we can represent translation as a matrix, we can compose it with other transformations

Example: translate 10 units down the Z then rotate 90° around X

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(90°) & -\sin(90°) & 0 \\ 0 & \sin(90°) & \cos(90°) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

why this is so beautiful ?

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$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

WHY THIS IS SO BEAUTIFUL ?

we can represent translation as a matrix, we can compose it with other transformations

Example: translate 10 units down the Z then rotate 90° around X

Caution: matrix multiplication is NOT commutative!

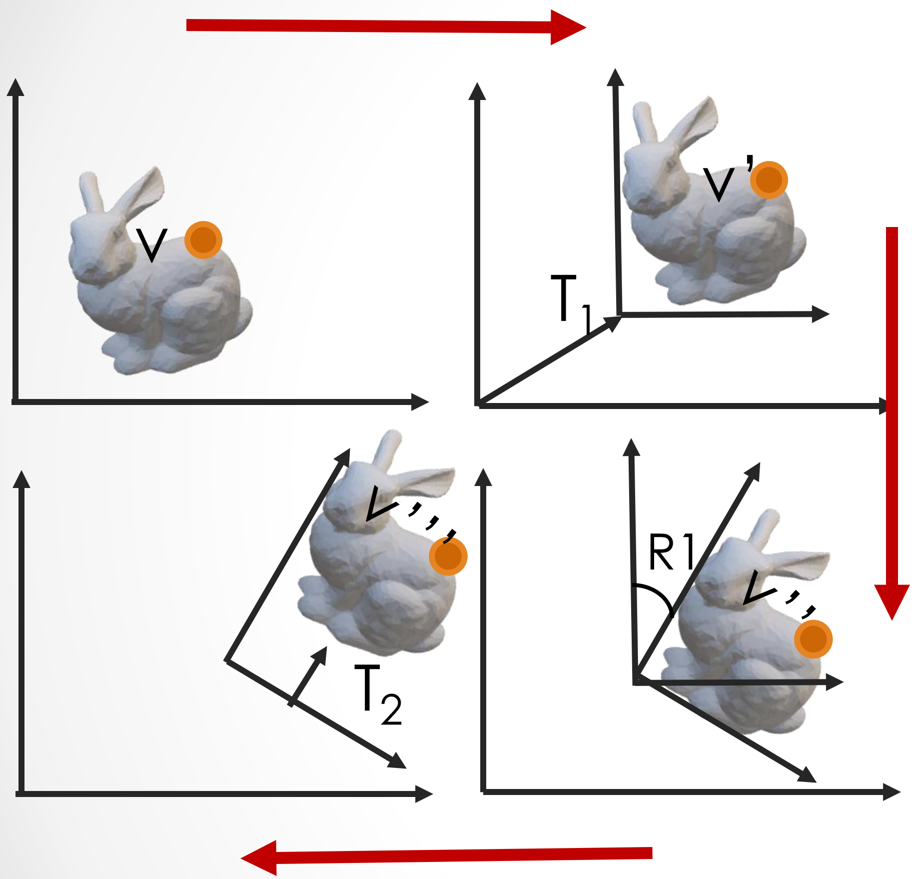
$$\begin{bmatrix} y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y \\ z \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

World Coordinates

- We have been discussing transformations as transforming points.
- Always need to think of the transformation in the world coordinate system.
- Useful to think of them as a change in coordinate system.
- Model objects in a local coordinate system, and transform into the world system.

World Coordinates



$$v' = T_1 v$$
$$v'' = T_1 R_1 v$$
$$v''' = T_1 R_1 T_2 v$$

- Concatenate local transformation matrices from left to right
- Can obtain the local world transformation matrix
- v',v'',v''' are the world coordinates of p after each transformation

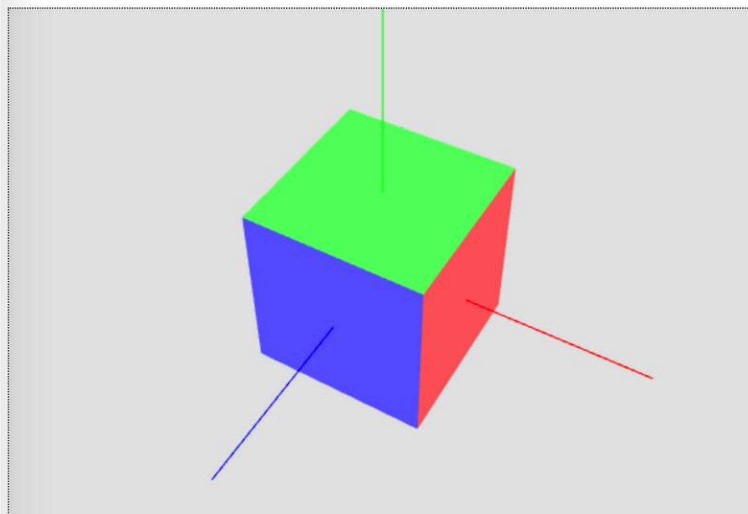
In class activity

In-class Activity (Group work)

3D Transformation

(1) Go to this website, where you will find an interactive demo of 3D transformations:

https://stevekautz.com/cs336f20/examples/transformations/RotatingCube.html



Keyboard controls:

SPACE - pause rotation
y - rotate about y axis
x - rotate about x axis

3 Building Blocks of 3D

Modelling

Lighting

Texturing

3 Building Blocks of 3D

Done

Lighting

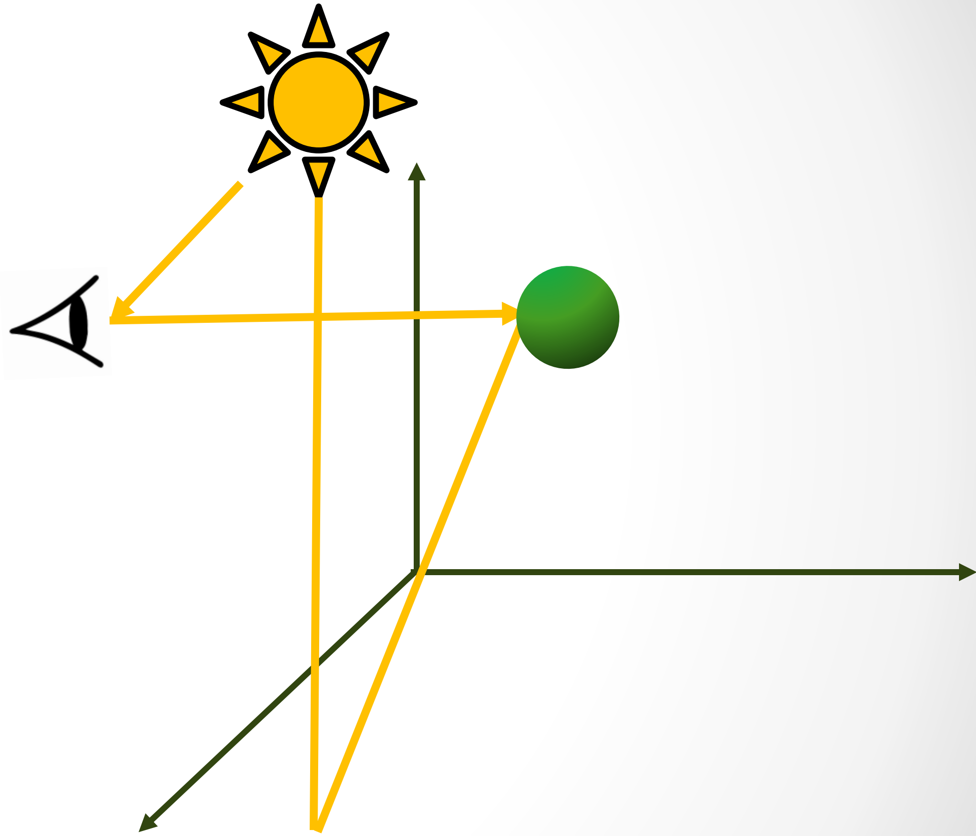
Texturing

Lighting

Lighting and Shading

lighting equation

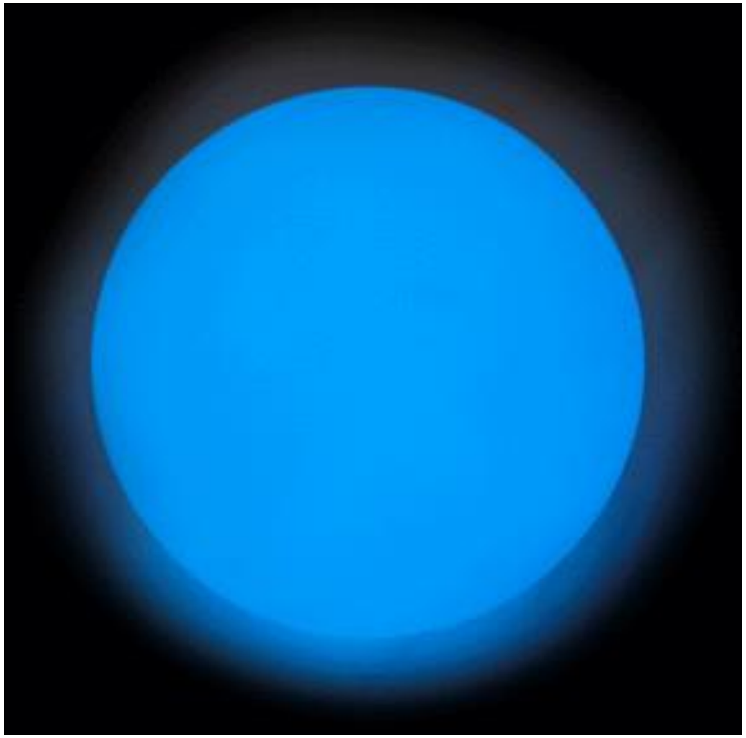
Emissive+
Ambient +
Diffuse +
Specular+



lighting Model

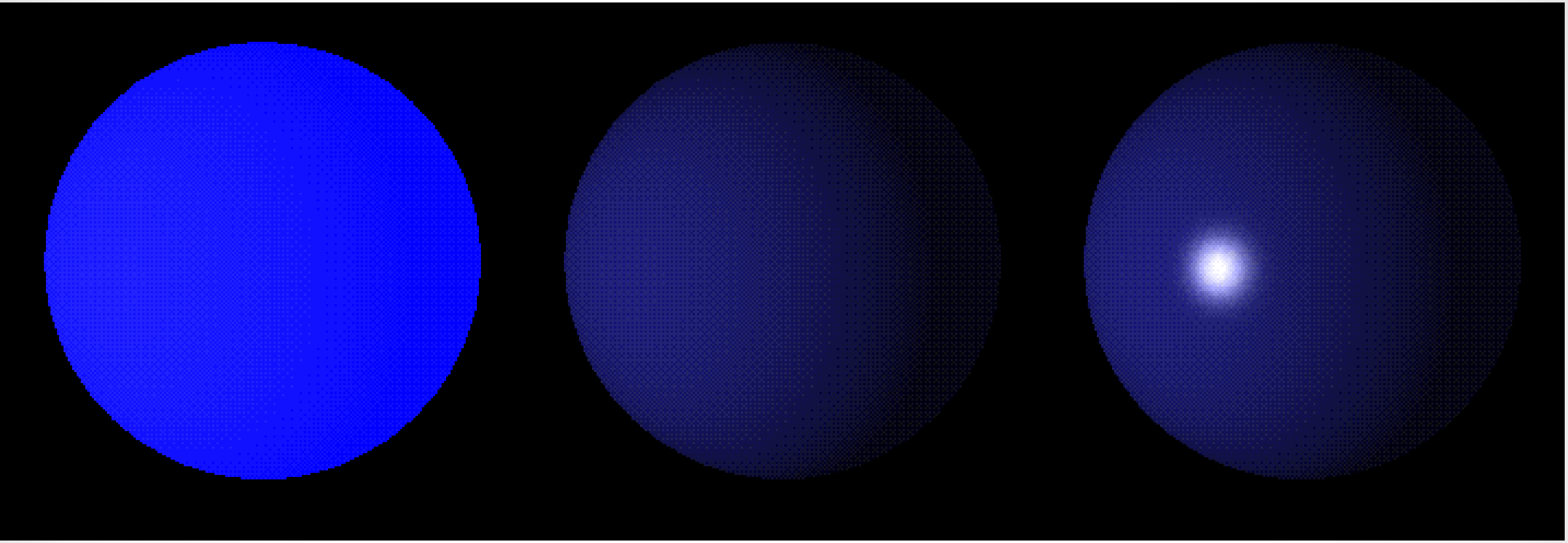
https://www.gabrielgambetta.com/computer-graphics-from-scratch/light.html

Lighting and Shading



Emissive

Lighting and Shading

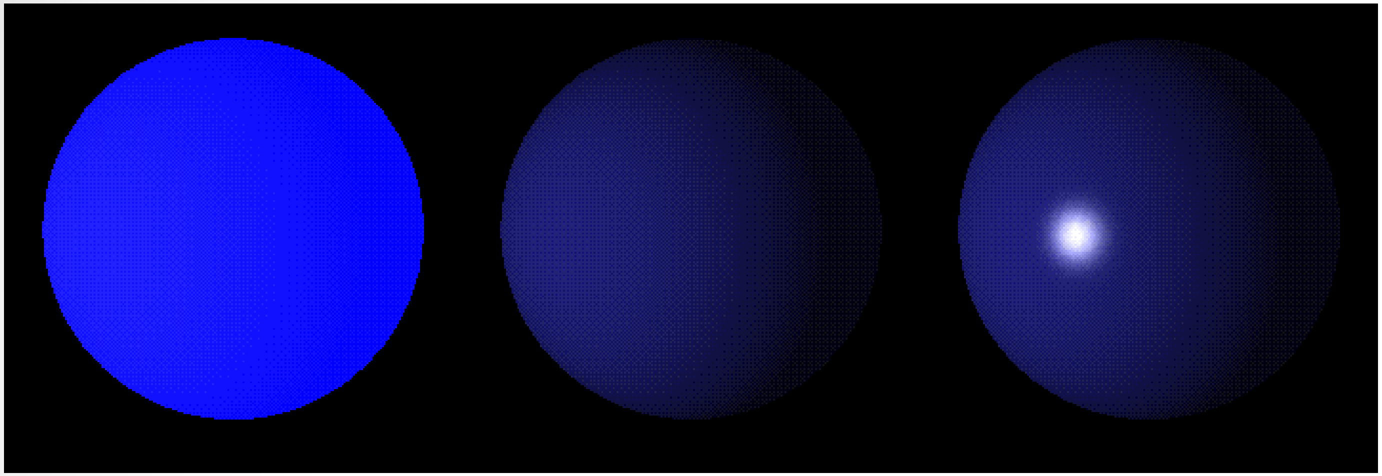


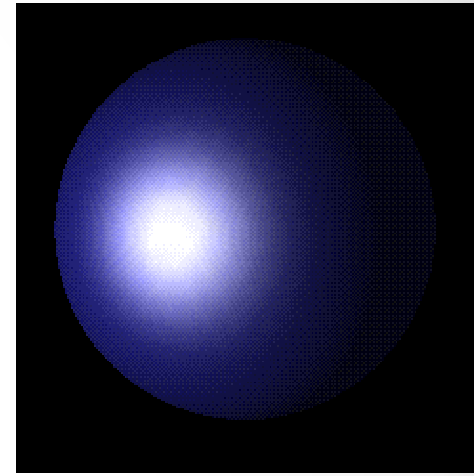
Ambient

Diffuse

Specular

Lighting and Shading





Ambient

Diffuse

Specular

Result

Lighting and Shading

lighting equation

The diffuse and specular components are computed based on the lights in the scene, while emissive and ambient are essentially independent.

The diffuse term can be thought of as a flat matte finish, and specular can be thought of as the shine finish of an object.

In terms of computation, the specular term is also affected by the camera angle.

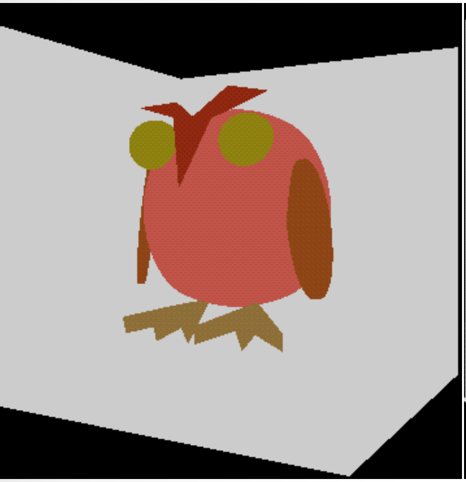
Lighting and Shading

Types of light sources

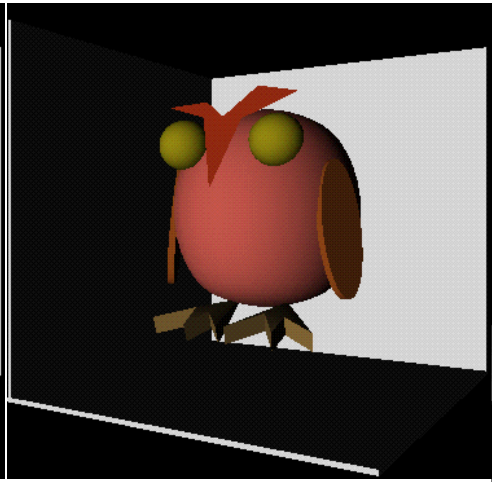
- **Ambient light:** no identifiable source or direction.
- **Point source:** given only by point.
- **Directional light:** given only by direction.
- **Spotlight:** from source in direction.
  - Cut-off angle defines a cone of light.
  - Attenuation function (brighter in centre).

Lighting and Shading

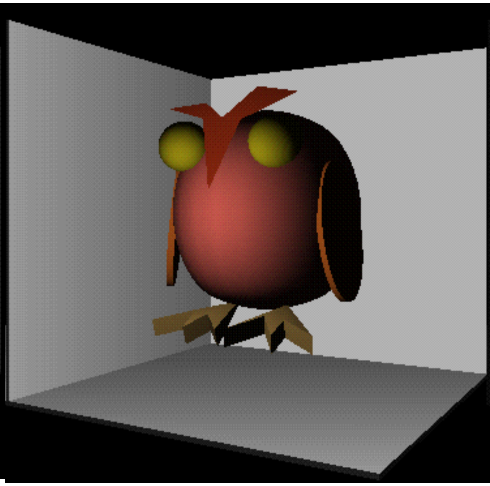
Types of light sources



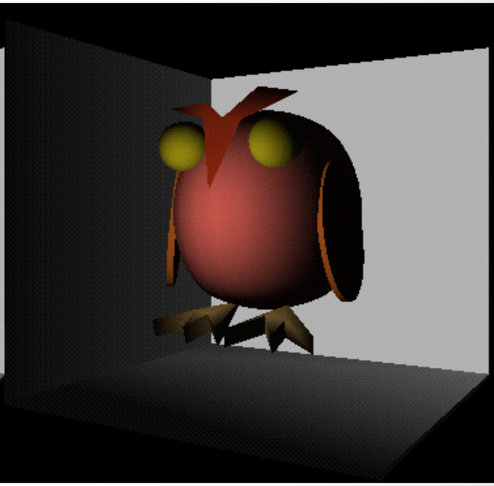
Ambient light



Directional light



Point source



Spotlight

Lighting and Shading

Shading Models

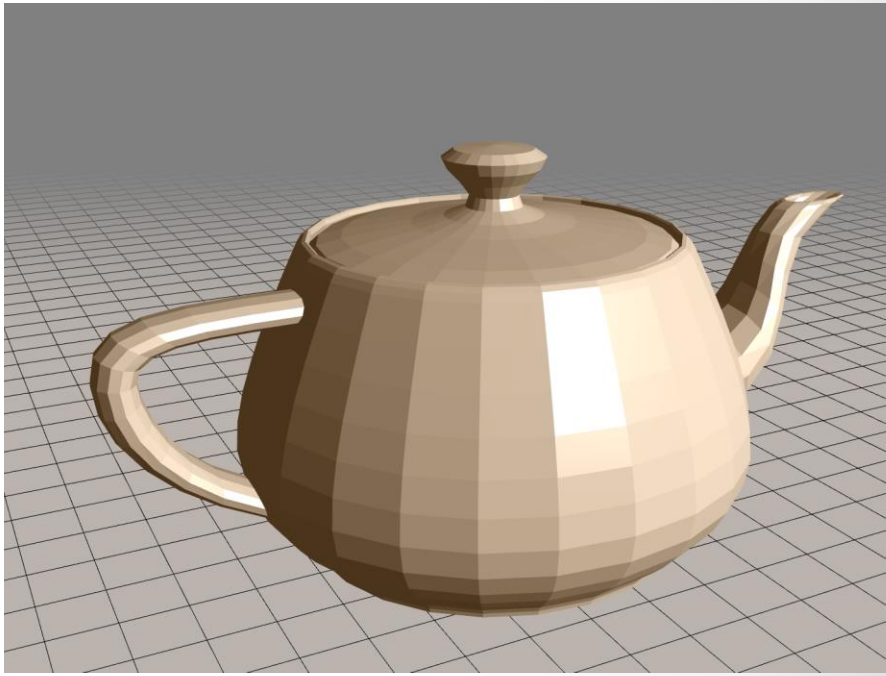
- Flat Model
- Gouraud Model
- Phong Model

Lighting and Shading

Flat Model

We define a normal at each polygon (not at each vertex)

Lighting: Evaluate the lighting equation at the center of each polygon using the associated normal



Shading: Each sample point on the polygon is given the interpolated lighting value at the vertices (i.e., a total hack)

Lighting and Shading

Gouraud Model

Define a normal vector at each *vertex*

Lighting: Evaluate lighting equation at each vertex using associated normal vector

Shading: Each sample point's color on polygon is interpolated from color values at polygon's vertices

An issue with Gouraud shading is that specular highlights appear to "jump" as the object rotates.

Lighting and Shading

Phong Model

Each vertex has an associated normal vector

Lighting: Evaluate lighting equation at each vertex using associated normal vector

Shading:

For every sample point on the polygon we interpolate normals at vertices of the polygon and compute color using lighting equation with the interpolated normal at each interior pixel

Lighting and Shading

Flat Model

Gouraud Model

Phong Model

In Class Activity

In-class Activity (Group work)

Shading Models

- A **shading model** is a mathematical formula to compute the surface characteristics. It determines how a surface is shaded. What is called a **shader** or **material** is an implementation of a shading model It is a technique that defines when and where we compute the components needed for **light** calculations.

- **Normals** play a central role in shading, it is known that an object becomes brighter if we orient it towards a light source. The orientation of an object surface plays an important role in the amount of light it reflects (and thus how bright it looks).

  - In 3D graphics, a **normal** is the word for a unit vector that describes the direction a surface is facing. The lines sticking out of the objects represent normals for each vertex (see the picture on the right). There are three main shading models that are used for different results: **flat** shading; **Gouraud** shading; **Phong** shading.

(1) Go to this website, where you will find an interactive demo of three type of shading models:

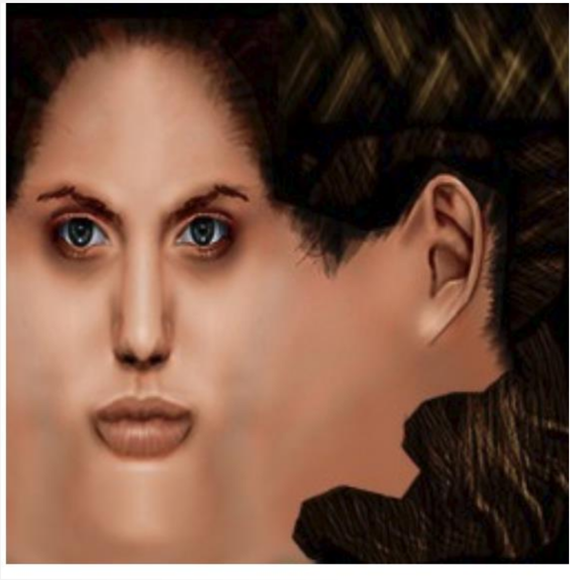
Flat, Phong, and Gouraud shading: http://www.realtimerendering.com/teapot/

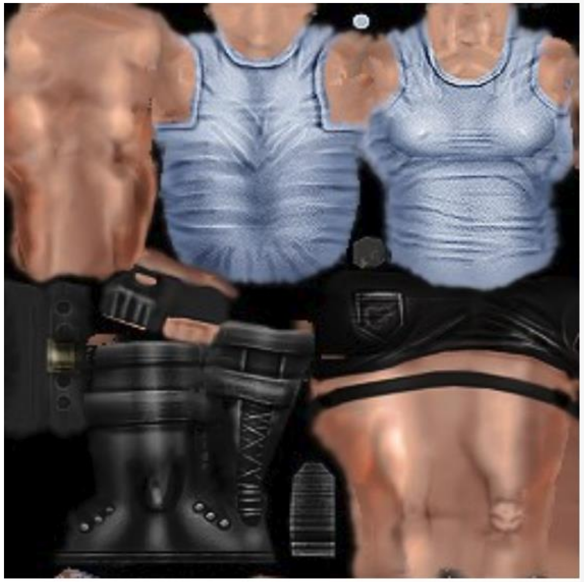
(2) Try to change the style of the shading models, and observe the difference in the results.

(3) Which shading model is illustrated in this picture?

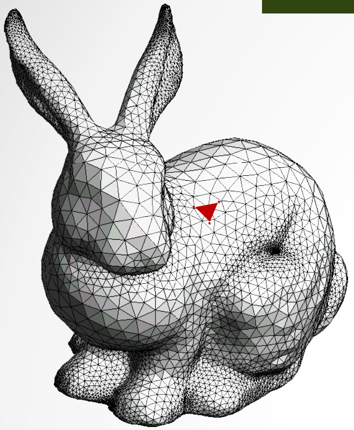
Texturing

Texturing

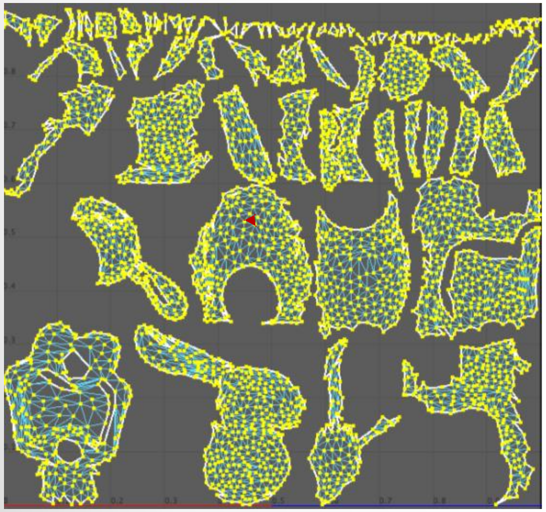




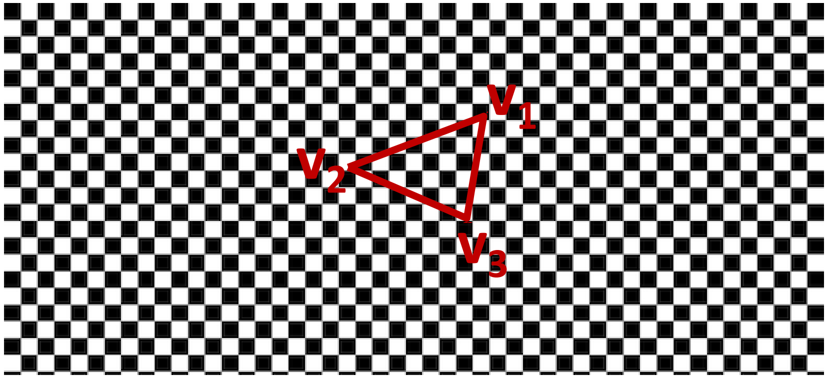
Texturing



geometry



- For each triangle in the model establish a corresponding region in the phototexture



2D image

Texture map

- During rasterization interpolate the coordinate indices into the texture map

3D Graphics Pipeline

Vertices

↓

Transformer — 3D model, scale, rotate, translate, lighting, shading, texturing

↓

Clipper — is it visible on screen?

↓

Projector — 3D to 2D

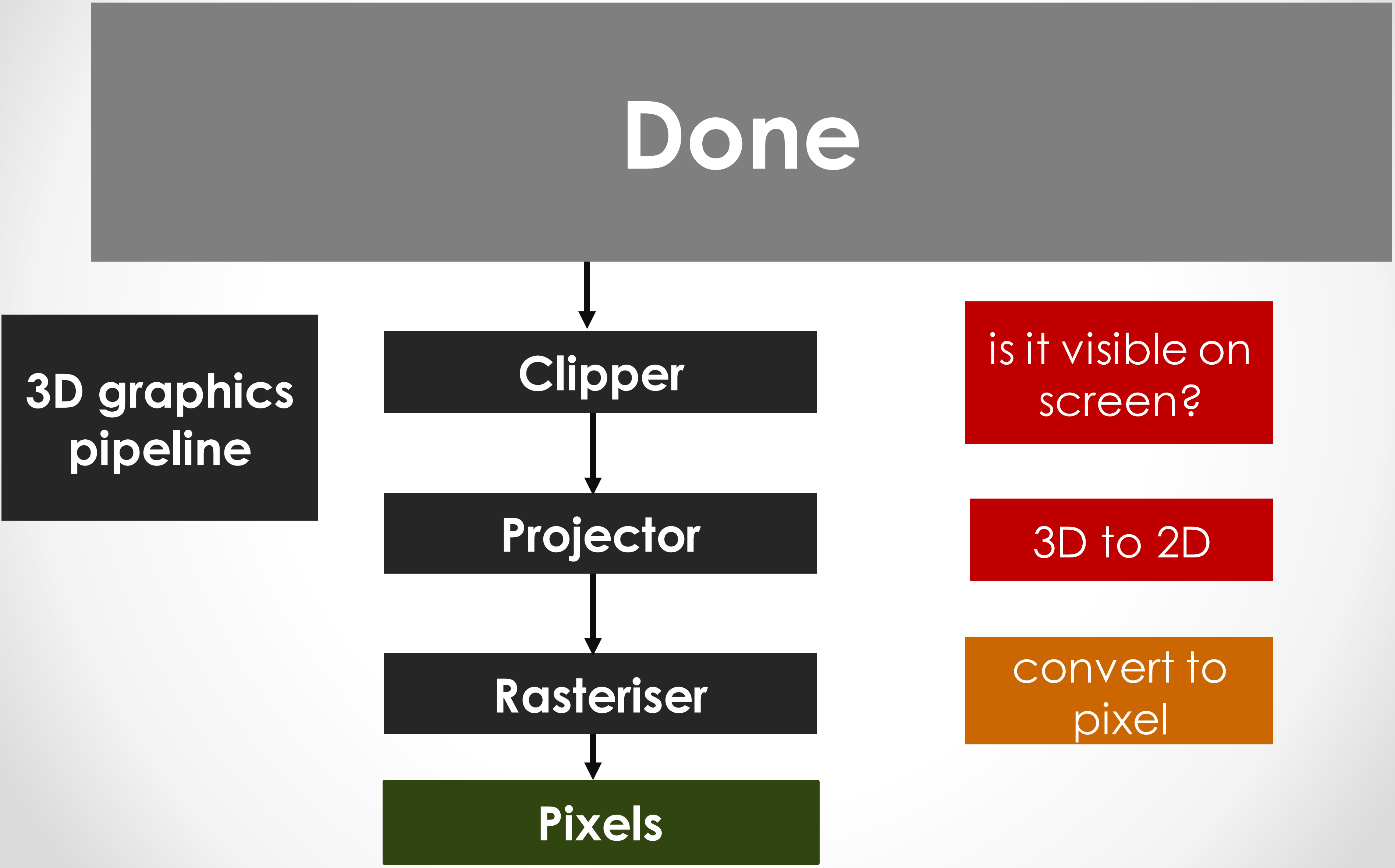
↓

Rasteriser — convert to pixel

↓

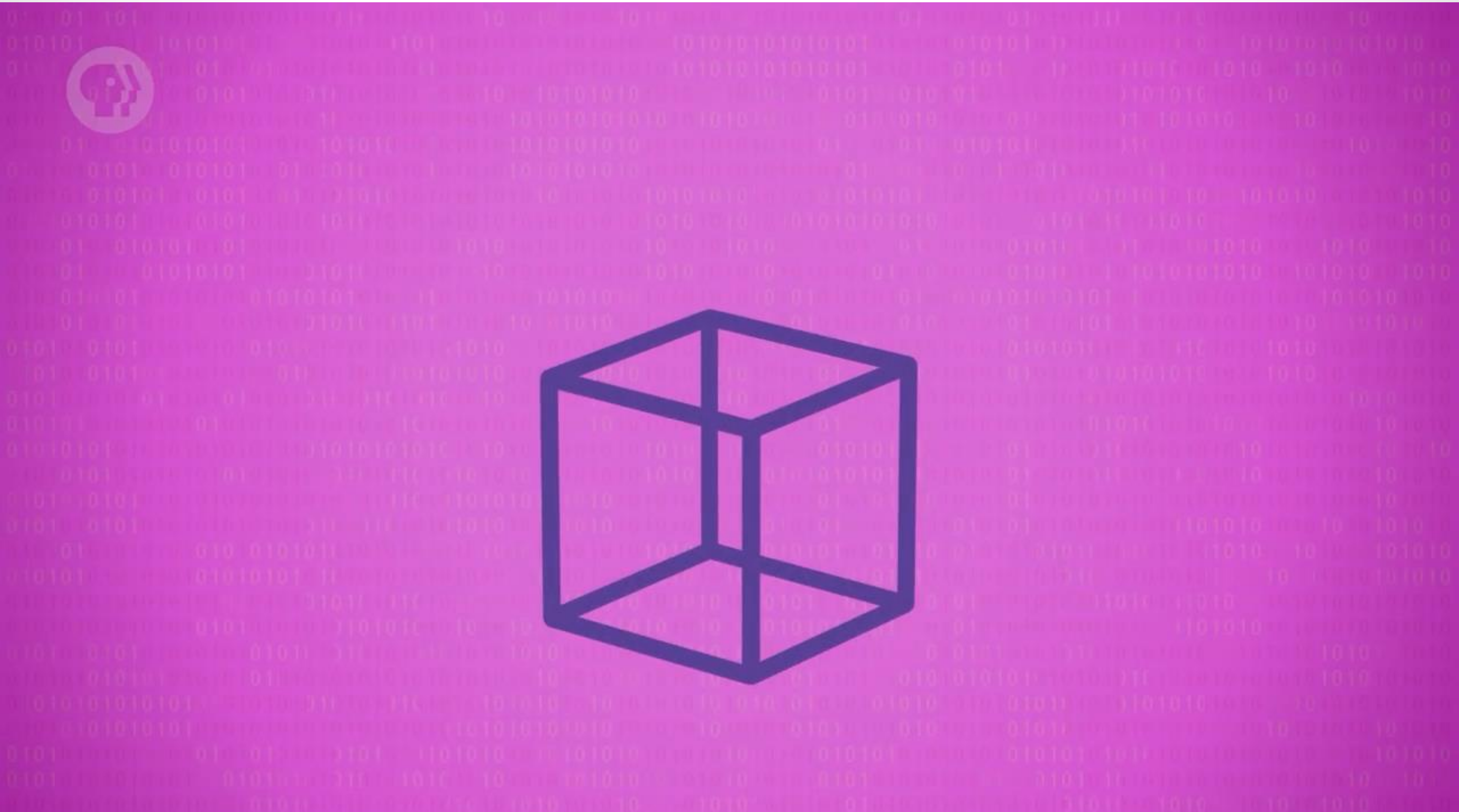
Pixels

Viewing and Projection



Viewing and Projection

Projection

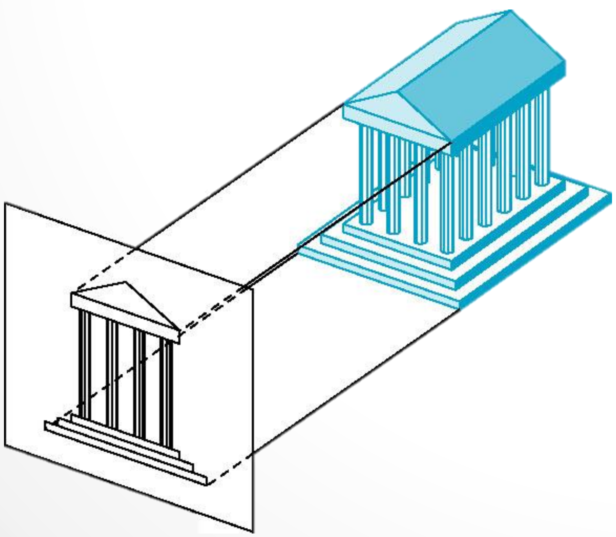


Viewing and Projection

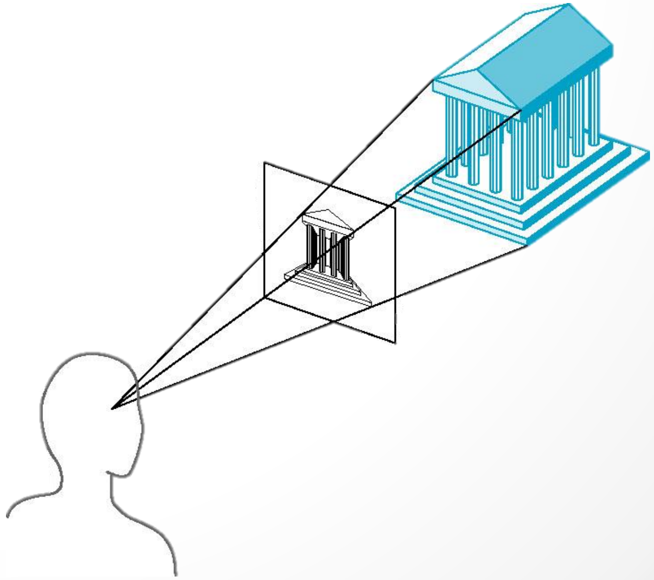
Projection

Complex transformation [Angel Ch. 5]

Orthographic Projection



Perspective Projection



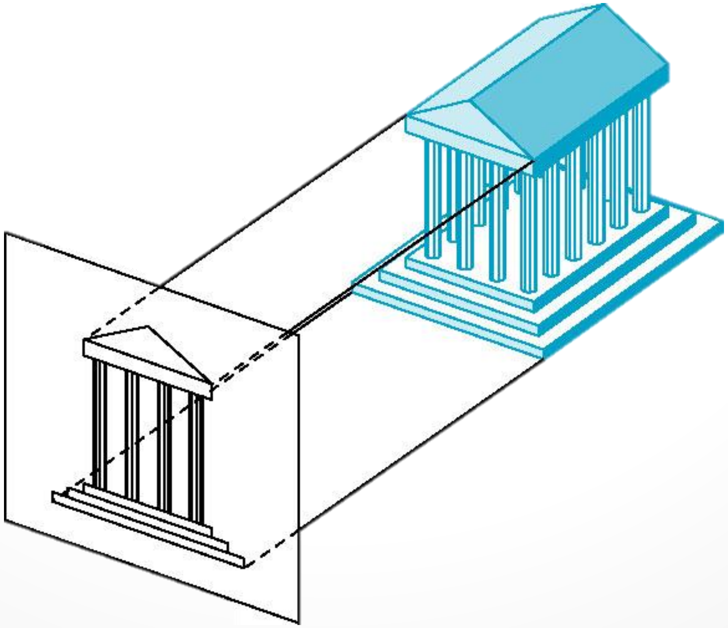
Viewing and Projection

Projection

Orthographic Projection

- **projection** that maintains parallel lines but provides no sense of depth.

- It is used in the engineering fields to create accurate renderings of models. They maintains parallel lines but provide no sense of depth. All vertices are projected straight onto a viewing window



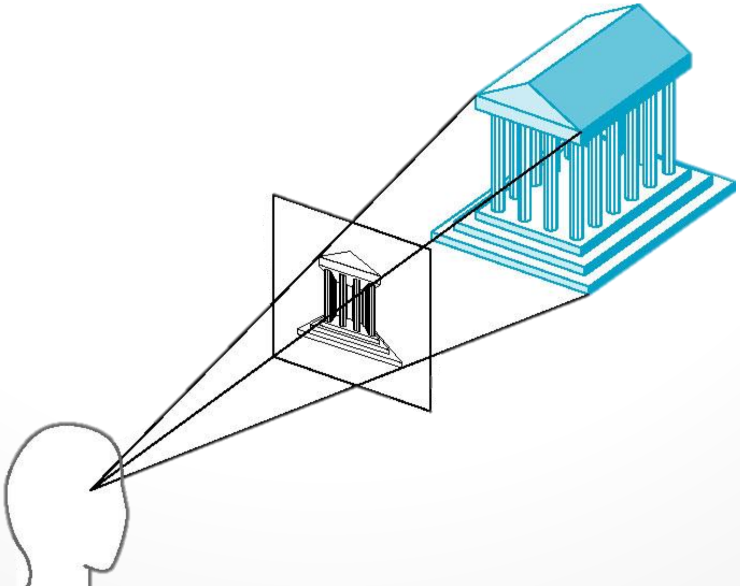
Viewing and Projection

Projection

Perspective Projection

- projection provides for a sense of depth, but parallel lines are skewed toward vanishing points

- a virtual scene to make it appear like a view from a real-world camera. Objects further from the camera appear to be smaller and all lines appear to project toward vanishing point which skew parallel lines. Perspective projections are almost always used in gaming, movie special effects, and visualizations of virtual worlds.

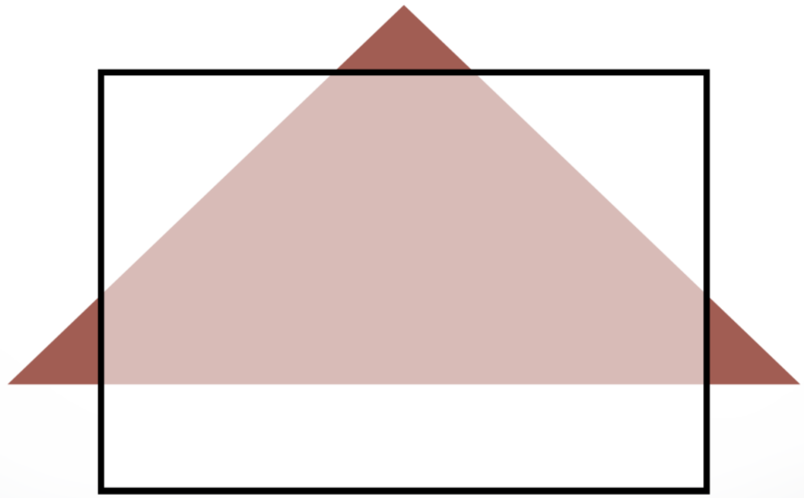


Viewing and Projection

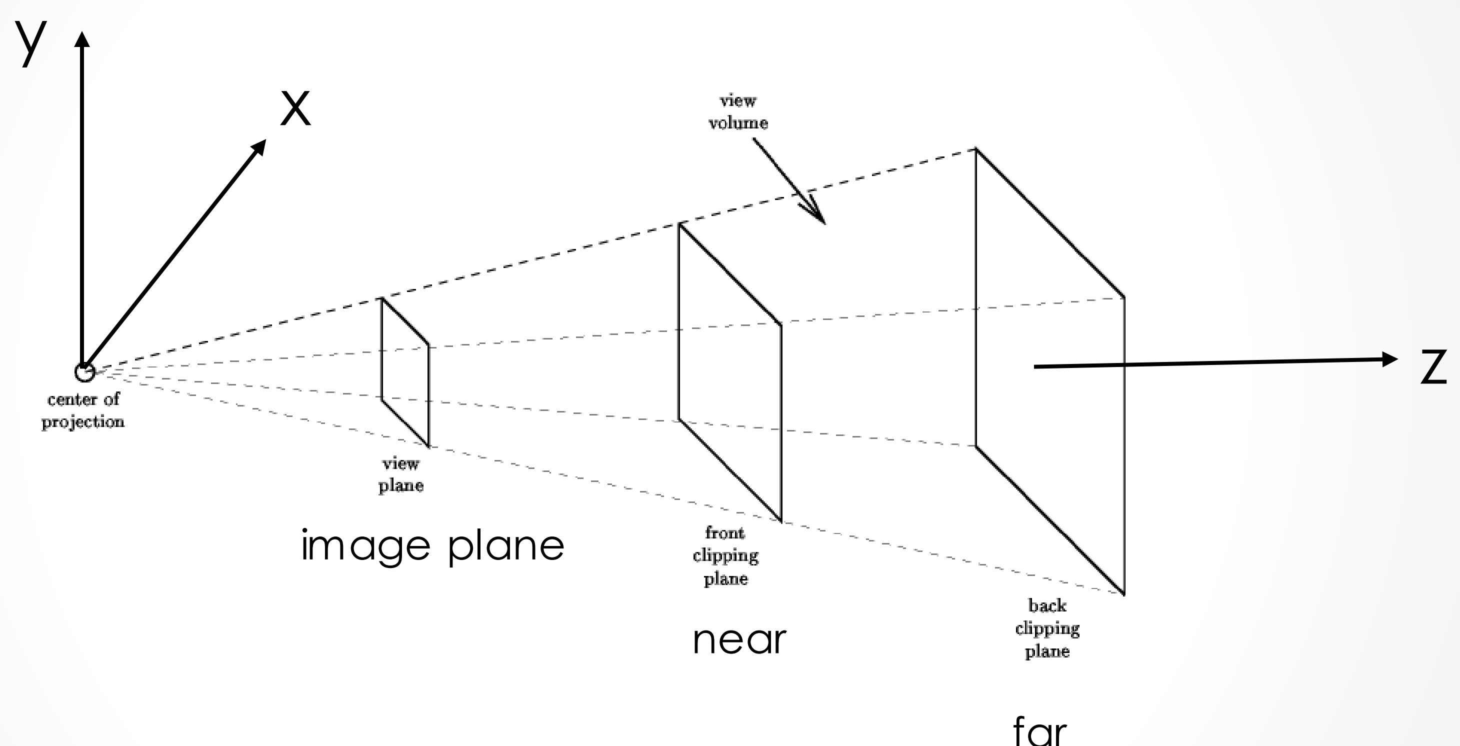
Clipping

Mostly automatic (must set viewport)

2D Clipping Problem



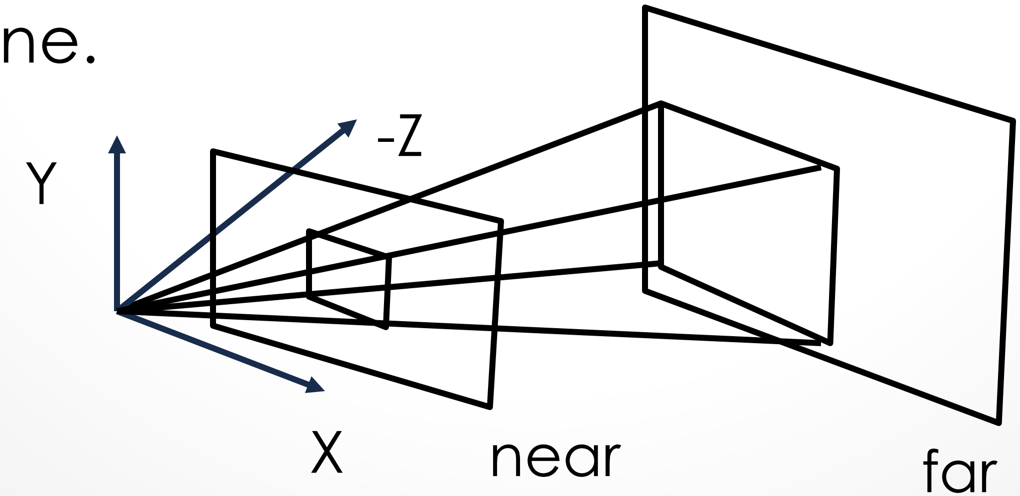
Viewing and Projection



Clipping is tricky because of frustum shape

Viewing and Projection

- Define the “view frustum” (6 parameters).
- Assume origin is the view point.
- Near and far planes (planes parallel to XY plane in the negative Z axis).
- Left, right, top, bottom rectangle defined on the near plane.



Rendering

is the automatic process of generating a photorealistic or non-photorealistic image from a 2D or 3D model

Rendering

The rendering equation is an integral equation based on light and is very computationally demanding

Aliasing

Rendering

The rendering equation is an integral equation based on light and is very computationally demanding

Aliasing

Aliasing: the process by which smooth curves or lines become jagged due to the resolution of the graphics device or insufficient data to represent the curves

Rendering

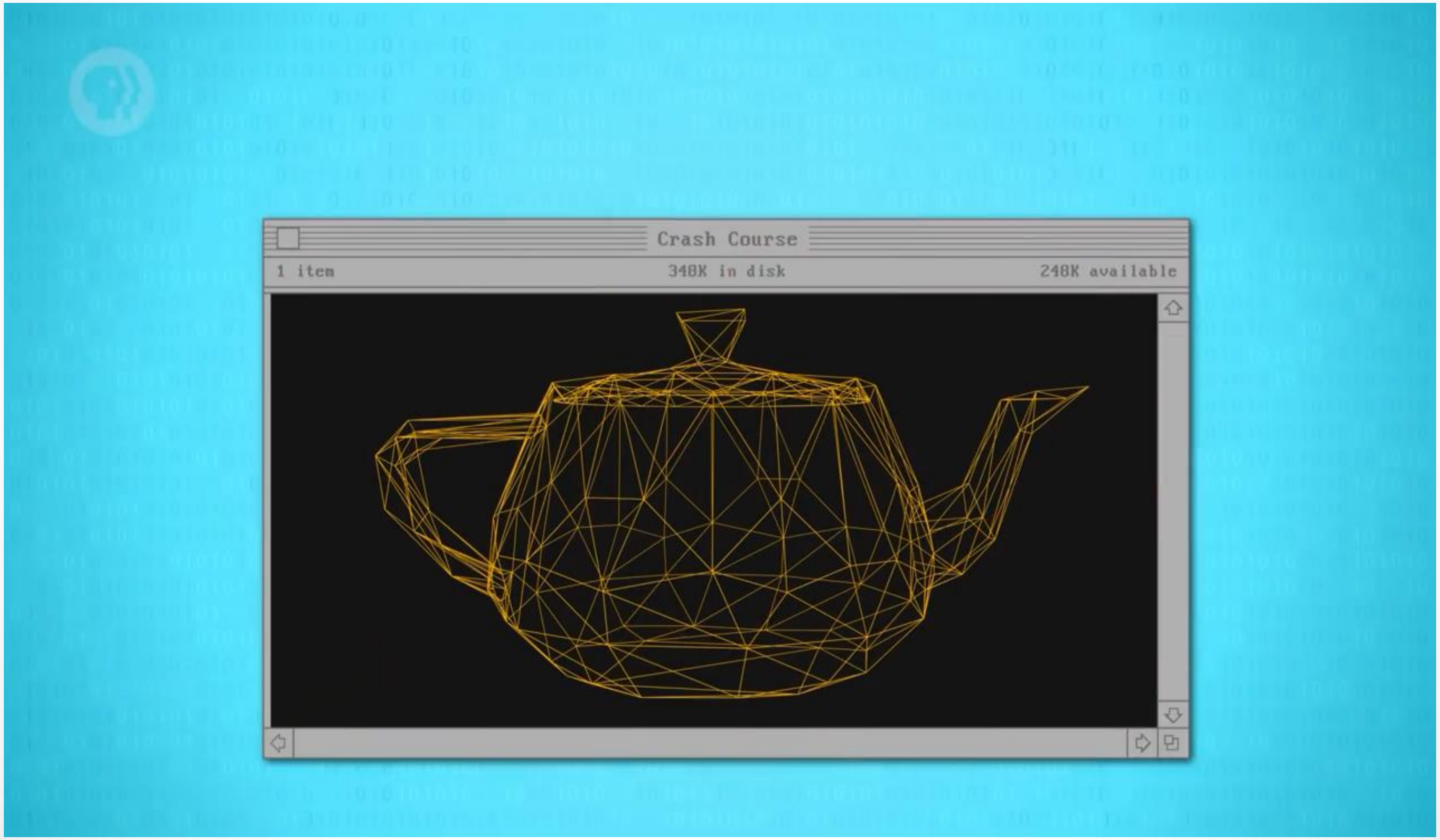
The rendering equation is an integral equation based on light and is very computationally demanding

Aliasing

Anti-aliasing is a technique for minimizing the distortion artefacts

Rendering

The rendering equation is an integral equation based on light and is very computationally demanding



https://www.youtube.com/watch?v=TEAtmCYYKZA

Questions?

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